Contents

Introduction **3**

Epigenetic Regulation by Non-Coding RNAs: The Temporal Dimension . . . 3

Control Theory and Biological Regulation . . . 4

The Second-Order Negative Feedback Paradigm . . . 4

Research Objectives and Approach . . . 5

Analytical Framework . . . 5

Methods **6**

The model . . . 6

Bode Plot . . . 10

Results **10**

Overview of Configurations Analyzed . . . 10

Configuration 1: Short miRNA Cascade (G-type) - Rapid Responsive Regulation . . . 11

Configuration 2: miRNA Cascade (H-type) - Enhanced Noise Rejection . . . 11

Configuration 3: Long ncRNA Cascade (G-type) - Ultra-Selective Temporal Integration . . . 13

Configuration 4: Mixed Cascade (miRNA + ncRNA) - Marginal Stability and Emergent Dynamics 13

Comparative Analysis: Architectural Principles . . . 14

Predicted Timescales and Experimental Validation . . . 15

Summary of Key Findings . . . 15

Discussion **16**

Summary of Key Findings . . . 16

Biological Significance of Frequency Filtering . . . 16

Architectural Design Principles . . . 16

Testable Predictions and Experimental Validation . . . 17

Application to Riboswitches . . . 17

Limitations and Future Directions . . . 19

Conclusions . . . 19

Appendix . . . 20

Declarations **22**

Consent to Participate declaration . . . 22

Ethics approval and consent to participate . . . 22

Consent for publication . . . 22

Availability of data and materials . . . 23

Competing interests . . . 23

Funding 23
Authors contributions 23
Acknowledgements 23

References **23**

From Sequence to Timescale: Frequency-Domain Analysis Reveals How ncRNA Length Determines Epigenetic Regulatory Dynamics

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Abstract

Background:Non-coding RNAs (ncRNAs) are central regulators of epigenomic states, orchestrating chromatin modifications, gene silencing, and nuclear architecture through recruitment of chromatin-modifying complexes. Long non-coding RNAs (lncRNAs) such as XIST, HOTAIR, and MALAT1 exemplify these regulatory roles. Despite their importance, quantitative computational frameworks for linking ncRNA sequence composition to temporal regulatory behavior remain limited.

We introduce a frequency-domain control theory framework modeling RNA sequences as cascaded feedback systems. Analysis reveals that RNA length and topology encode low-pass temporal filtering properties, with ncRNAs exhibiting slower cutoff frequencies and stronger noise attenuation than short miRNAs. Mixed miRNA–ncRNA cascades show marginal stability suggestive of bistable epigenetic regulation.

Conclusions: ncRNA sequence composition systematically determines temporal response characteristics relevant to epigenetic regulation, enabling predictive modeling of chromatin dynamics and RNA-mediated control systems.

Long non-coding RNAs (lncRNAs) are central regulators of epigenomic states, orchestrating chromatin modifications, gene silencing, and nuclear architecture through recruitment of chromatin-modifying complexes. RNAs such as XIST, HOTAIR, MALAT1, and NEAT1 control critical epigenetic processes including X-chromosome inactivation, Polycomb-mediated silencing, alternative splicing regulation, and paraspeckle formation. Despite their importance in epigenetic regulation, we lack quantitative computational frameworks for characterizing how ncRNA sequence composition determines temporal response properties. Understanding these dynamics is essential for predicting regulatory timescales, noise filtering capacity, and responsiveness to developmental or environmental signals that trigger epigenomic remodeling. Current computational approaches focus predominantly on static sequence features that cannot predict dynamic behaviors such as response latency or frequency-selective signal filtering.

Methods We present a computational framework applying frequency-domain control theory to characterize temporal dynamics of epigenetically active ncRNAs based on sequence composition. RNA sequences are modeled as cascades of second-order negative feedback systems, where each nucleotide contributes a transfer function stage with parameters derived from molecular properties or experimentally determined kinetic rates. Four regulatory configurations were analyzed: feed-forward miRNA cascades, feedback miRNA cascades, long chromatin-regulatory ncRNAs, and mixed miRNA–ncRNA cascades modeling competitive endogenous RNA networks. System behavior was evaluated using Bode plot analysis across six orders of magnitude in frequency.

Results Short miRNA cascades exhibited cutoff frequencies of approximately 0.012 rad/s, corresponding to response periods of 8–9 minutes, with moderate DC gains. Long ncRNA cascades showed lower cutoff frequencies (approximately 0.0048 rad/s) and substantially higher DC gains, indicating strong temporal integration. Feedback architectures demonstrated increased phase lag and enhanced high-frequency attenuation relative to feed-forward configurations. Mixed cascades exhibited marginal stability with infinite DC gain, suggesting potential bistable or oscillatory behavior. Application of the framework to metabolite-sensing

riboswitches yielded sub-second response timescales, supporting generalizability across RNA regulatory systems.

Conclusions RNA regulatory cascades act as tunable low-pass temporal filters whose properties are encoded by sequence length and network topology. The predicted timescales align with experimental observations of chromatin remodeling, transcriptional bursting, and metabolic sensing. ncRNAs appear optimized for noise filtering and stable epigenetic regulation, whereas short miRNAs enable rapid and flexible control. This framework provides a quantitative basis for interpreting time-resolved epigenomic data and for engineering synthetic RNA-based chromatin regulators with defined temporal behavior. —

Keywords: Non-coding RNA, ncRNA length, Epigenetic regulation, Chromatin dynamics, Control theory, Frequency-domain analysis, Temporal filtering, Transfer functions, Bode plot analysis, RNA regulatory networks, miRNA, lncRNA

Introduction

Epigenetic Regulation by Non-Coding RNAs: The Temporal Dimension

Epigenetic modifications—including DNA methylation, histone post-translational modifications, and chromatin remodeling—govern gene expression programs without altering DNA sequence. These modifications are essential for development, cellular differentiation, environmental adaptation, and disease progression. Non-coding RNAs (ncRNAs) particulary long non-coding RNAs (lncRNAs) have emerged as master regulators of epigenomic states, functioning as molecular scaffolds that recruit chromatin-modifying complexes to specific genomic loci (Brown et al. 1992; Rinn et al. 2007; Tripathi et al. 2010). The paradigmatic example is XIST (~17,000 nucleotides), which initiates X-chromosome inactivation by coating the inactive X and recruiting Polycomb Repressive Complex 2 (PRC2) to establish chromosome-wide silencing through H3K27me3 (Cifuentes-Rojas et al. 2014) deposition (Brown et al. 1992). Similarly, HOTAIR (~2,200 nucleotides) guides PRC2 to HOXD loci for transcriptional repression during development (Rinn et al. 2007), MALAT1 (~8,000 nucleotides) modulates alternative splicing through regulation of SR splicing factor phosphorylation and chromatin accessibility (Tripathi et al. 2010), and NEAT1 (~23,000 nucleotides) organizes paraspeckle nuclear bodies that sequester chromatin regulators and control gene expression in response to cellular stress. Despite extensive characterization of lncRNA-protein interactions, chromatin binding patterns, and regulatory targets through ChIP-seq and RNA-seq approaches, a fundamental question remains poorly addressed by computational methods: how does ncRNA sequence composition determine temporal response properties in epigenetic regulation? Chromatin modifications occur on characteristic timescales spanning seconds to hours—histone acetylation and deacetylation occur within minutes, while DNA methylation establishment and heterochromatin formation require tens of minutes to hours. LncRNAs operating within these systems must respond appropriately to transient versus sustained regulatory signals, filter molecular noise arising from stochastic transcription, and coordinate timing across multiple sequential regulatory steps. Yet current computational approaches in epigenomics focus predominantly on static sequence features—binding motifs for chromatin regulators, secondary structure prediction, RNA-protein interaction sites, and network topology— that cannot predict dynamic behaviors such as response latency, signal integration timescales, frequency- selective filtering, or temporal coordination with chromatin-modifying enzymes. Understanding the relationship between ncRNA sequence and temporal dynamics is essential for three interconnected challenges in computational epigenomics: (1) deciphering natural regulatory mechanisms that control chromatin state transitions, (2) predicting how ncRNAs integrate noisy transcriptional signals to produce stable epigenetic outputs, and (3) engineering synthetic RNA circuits for targeted epigenetic manipulation with predictable temporal control. Temporal Dynamics in Epigenomic Regulation: Epigenetic processes exhibit characteristic timescales that must be matched by their regulatory RNAs for functional coordination: X-chromosome inactivation: XIST spreading across the X chromosome occurs over 30-60 minutes during early development (Engreitz et al. 2013), with initial PRC2 recruitment and H3K27me3 deposition beginning within 10-15 minutes of XIST upregulation. This progressive silencing requires temporal integration—XIST must distinguish transient transcriptional fluctuations from sustained developmental signals that commit to

inactivation. Polycomb-mediated gene silencing: Following lncRNA-mediated recruitment of PRC2 to target loci, H3K27me3 deposition exhibits lag times of 15-40 minutes. Chromatin compaction and gene silencing emerge over 1-2 hours. These multi-step processes require regulatory RNAs to maintain sustained signaling despite stochastic variations in protein concentrations. Chromatin remodeling (Narlikar, Sundaramoorthy, and Owen-Hughes 2013; Clapier et al. 2017)dynamics: ATP-dependent chromatin remodelers (SWI/SNF, ISWI, CHD families) recruited by lncRNAs show nucleosome repositioning timescales of 5-20 minutes. Coordinated remodeling across gene loci requires lncRNAs to integrate signals over these characteristic periods. Transcriptional memory and epigenetic inheritance: Epigenetic marks persist for 1-6 hours following removal of inducing signals, maintaining transcriptional states through cell divisions. Regulatory lncRNAs must filter rapid signal fluctuations to establish stable chromatin modifications that encode cellular memory. Nuclear body dynamics: NEAT1-dependent paraspeckle assembly and disassembly occur over 20-60 minutes in response to cellular stress, requiring coordinated recruitment of dozens of protein components. The temporal integration properties of NEAT1 determine stress response kinetics. Stochastic transcription and noise filtering: Transcriptional bursts—rapid pulses of mRNA production occurring on 5-15 minute timescales—generate substantial molecular noise. LncRNAs regulating chromatin must filter this noise to prevent spurious epigenetic modifications while responding to genuine sustained regulatory signals. These diverse timescales pose a fundamental question: do ncRNA sequence properties encode temporal filtering characteristics that match the kinetics of chromatin-modifying processes? Specifically, how does sequence length from short miRNAs ($\sim$ 22 nt) to long lncRNA (>200 nt) determine temporal response characteristics?

Control Theory and Biological Regulation

Biological systems universally employ feedback control to maintain homeostasis, reject disturbances, and respond appropriately to environmental signals (Del Vecchio and Murray 2014). Control theory—the mathematical framework for analyzing feedback systems—has been successfully applied to diverse physiological processes including hormonal regulation, metabolic pathways, neural circuits, cell cycle control, and immune responses. A hallmark of control-theoretic analysis is frequency-domain characterization, which reveals how systems amplify or attenuate signals at different temporal frequencies. This perspective is particularly powerful for understanding regulatory networks where timescale separation, noise filtering, and oscillatory behavior play functional roles. In the context of epigenetic regulation, frequency-domain analysis addresses questions such as: At what timescales do lncRNAs respond most strongly? How effectively do they reject high-frequency noise from transcriptional bursting? Do different lncRNA architectures (short vs. long, different topologies) confer distinct temporal filtering properties? Can interacting lncRNA networks generate bistable switches or oscillations characteristic of cell fate decisions? Despite the success of control theory in systems biology, its application to RNA-mediated epigenetic regulation has been limited. Existing computational models of chromatin regulation typically treat ncRNAs as discrete switches or static interaction nodes, neglecting their intrinsic dynamic properties. Yet ncRNAs are inherently dynamical systems—they fold, unfold, bind protein partners, undergo conformational changes, and interact with chromatin, all on characteristic timescales that determine their epigenetic regulatory function. We hypothesize that ncRNAs can be profitably modeled as feedback control systems whose dynamic properties are encoded in sequence composition, enabling quantitative prediction of temporal behaviors from sequence alone.

The Second-Order Negative Feedback Paradigm

We propose that RNA regulatory networks in epigenetic systems operate as cascades of second-order negative feedback systems—the canonical servo model extensively validated in control engineering (Ogata 2010). This choice is motivated by both mathematical parsimony and biological necessity: Mathematical rationale: Second-order systems represent the minimal framework capable of capturing essential regulatory dynamics—response speed (natural frequency ω_n) and stability characteristics (damping ratio ζ. First-order models cannot generate oscillations, overshooting, or resonance behaviors observed in many epigenetic processes (e.g., oscillatory gene expression, bistable cell fate switches). Higher-order models introduce unnecessary complexity without corresponding mechanistic justification when modeling individual nucleotide contributions.

Biological rationale: Viable epigenetic regulatory systems must exhibit negative feedback to maintain stability and robustness against perturbations. Without feedback, chromatin states would drift uncontrollably in response to stochastic fluctuations in transcription factor concentrations, chromatin modifying enzyme levels, or environmental signals. The second-order negative feedback architecture ensures bounded responses, finite settling times, and tunable trade-offs between speed and stability—all essential properties of functional epigenetic regulation that must persist through cell divisions and developmental transitions. This perspective reframes lncRNA sequences not as passive scaffolds that simply bind proteins, but as active temporal filters that selectively respond to chromatin regulatory signals based on their frequency content— analogous to how electronic filters in signal processing selectively pass or reject signals at different frequencies.

Research Objectives and Approach

In this work, we develop a frequency-domain computational framework that quantitatively links RNA sequence composition to dynamic response properties relevant to epigenetic regulation. Our approach has three key components: 1. Sequence-to-parameter mapping: We derive control-theoretic parameters (natural frequency ω_n, damping factor ζ directly from molecular properties. For sequence-based analysis, ω_n is scaled by nucleotide molecular weight (Adenine: 135 Da, Guanine: 151 Da, Cytosine: 111 Da, Uracil: 112 Da), reflecting how molecular mass influences conformational dynamics and binding kinetics. For systems with experimental data (e.g., riboswitches with measured binding and folding rates), parameters are computed from kinetic rate constants. 2. Cascade transfer functions: RNA sequences are modeled as cascades where each nucleotide contributes a second-order transfer function stage. We investigate two cascade architectures: G-type (feed-forward): Sequential signal propagation with phase lead compensation, modeling rapid responsive regulation appropriate for epigenetic establishment during developmental transitions H-type (feedback): Enhanced feedback with greater damping, modeling stable maintenance regulation appropriate for epigenetic memory and inheritance (Ragunathan, Jih, and Moazed 2015). ## Frequency response analysis: We use Bode plot analysis—the standard tool for characterizing dynamic systems—to extract quantitative metrics including cutoff frequencies (determining response timescales), bandwidth (determining signal selectivity), gain characteristics (determining sensitivity), and phase relationships (determining coordination with downstream processes). These metrics directly translate to epigenetically relevant properties: cutoff frequency determines how quickly lncRNAs respond to chromatin regulatory signals, bandwidth determines ability to distinguish sustained signals from transcriptional noise, and DC gain determines steady-state sensitivity to persistent inputs.

Analytical Framework

We analyze four representative configurations to systematically investigate effects of sequence length, cascade architecture, and regulatory interactions on temporal dynamics: 1. 2. 3. 4. Configuration 1: 22-nucleotide miRNA sequence with G-type topology (feed-forward cascades) - representing rapid responsive regulation Configuration 2: 22-nucleotide miRNA sequence with H-type topology (feedback cascades) - representing stable maintenance regulation Configuration 3: 140-nucleotide lncRNA sequence with G-type topology - representing chromatin- regulatory long ncRNAs Configuration 4: Mixed cascade combining miRNA (G-type) and lncRNA (H-type) in series - modeling competitive endogenous RNA networks These configurations enable direct comparison of: Architectural effects: G versus H topology while holding sequence constant (Configs 1 vs. 2) Length effects: Short versus long sequences while holding topology constant (Configs 1 vs. 3), representing miRNA versus chromatin-regulatory lncRNA dynamics Interaction effects: Combined regulatory networks (Config 4) modeling ceRNA crosstalk in epigenetic regulation For each configuration, we compute Bode plots across six decades of frequency (10^{-3} to 10^{3} rad/s, corresponding to timescales from ~1 second to ~17 minutes) and extract performance metrics. To demonstrate broader applicability beyond epigenetic regulatory lncRNAs, we also analyze five metabolite-sensing riboswitches using experimentally determined kinetic parameters, validating the framework's predictive capacity. ## Key Questions Addressed {-}

This computational framework allows us to address fundamental questions in epigenomic regulation: What are the characteristic timescales of lncRNA-mediated chromatin regulatory responses predicted by sequence composition? How do these predictions compare to experimentally observed dynamics of chromatin modification, XIST spreading, Polycomb recruitment, and transcriptional memory? How does sequence length affect temporal filtering properties and noise rejection capacity—specifically, do long lncRNAs filter transcriptional bursting more effectively than short miRNAs?(Suter et al. 2011; Larsson et al. 2019), What functional differences arise from alternative cascade architectures (feed-forward versus feedback), and do these correspond to distinct epigenetic regulatory strategies (establishment versus maintenance)? Can miRNA-lncRNA interactions in ceRNA networks create emergent dynamical behaviors such as bistability or oscillations relevant to cell fate decisions? Does the framework provide quantitative insights into riboswitch robustness and metabolic sensing dynamics, validating its applicability beyond epigenetic contexts? By answering these questions through rigorous frequency-domain analysis, we aim to establish a quantitative computational foundation for understanding how ncRNA sequence composition determines temporal dynamics in epigenetic regulation, with implications for interpreting time-series epigenomic data, predicting lncRNA function from sequence, and engineering synthetic RNA circuits for targeted chromatin modification.

Methods

The model

The model is a variant of a previously published second order negative feedback model for description of plasma/blood kinetics of substrates and hormones shown in Figure 1 (Boroujerdi and Schmidt 2013; BOROUJERDI 2024)

The differential equations describing the model in Figure 1 are:

$$\frac{\mathbf{ds}}{\mathbf{dt}} = \mathbf{Ra} - \mathbf{k_3C} \tag{1}$$

$$\frac{\mathbf{dC}}{\mathbf{dt}} = \mathbf{k_4S} - \mathbf{k_2C} \tag{2}$$

At steady state condition that is, when left hand sides of equations (1) and (2) are zero and substituting for C from (2) into (1)

$$\mathbf{Ra} = \frac{\mathbf{k_3k_4}}{\mathbf{k_2}}\mathbf{S} \tag{3}$$

Where S is a label and can represent a substrate, hormone, ligand or any other biological entity with concentration. At initial condition, e.g. $S(0)$ or at any discrete time value t_i that the system has reached, e.g. $S(t_i)$ following a change in its production rate $Ra(t_i)$. Note that we are assuming that the system has reached a steady state. The coefficient of equation (3) is termed the clearance rate (cr) with dimension of $/time$. Note that this formulation uses conservation of mass for each of the compartments S and C. The label C represent controller of S where all controlling action takes place. This model also represent a servo and as such would respond to it its input. For example, if it is required that S reaching a value b from its current position the Ra would be crb. The system can also be presented using the state space formulation for linear dynamical systems, that is,

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu} \tag{4}$$

$$\mathbf{y} = \mathbf{Cx} \tag{5}$$

where A, B and C are matrices with dimensions n x n, n x r and m x n respectively. It is noted that for the state space representation of the model shown in Figure 1 the magnitude of u is given by equation (3). It can be shown that, the parameters of the model can be related to the standard second order differential equation parameterised by ω_n and ζ with $k_2 = 2\zeta\omega_n$ and $k_3 = k_4 = \omega_n$.

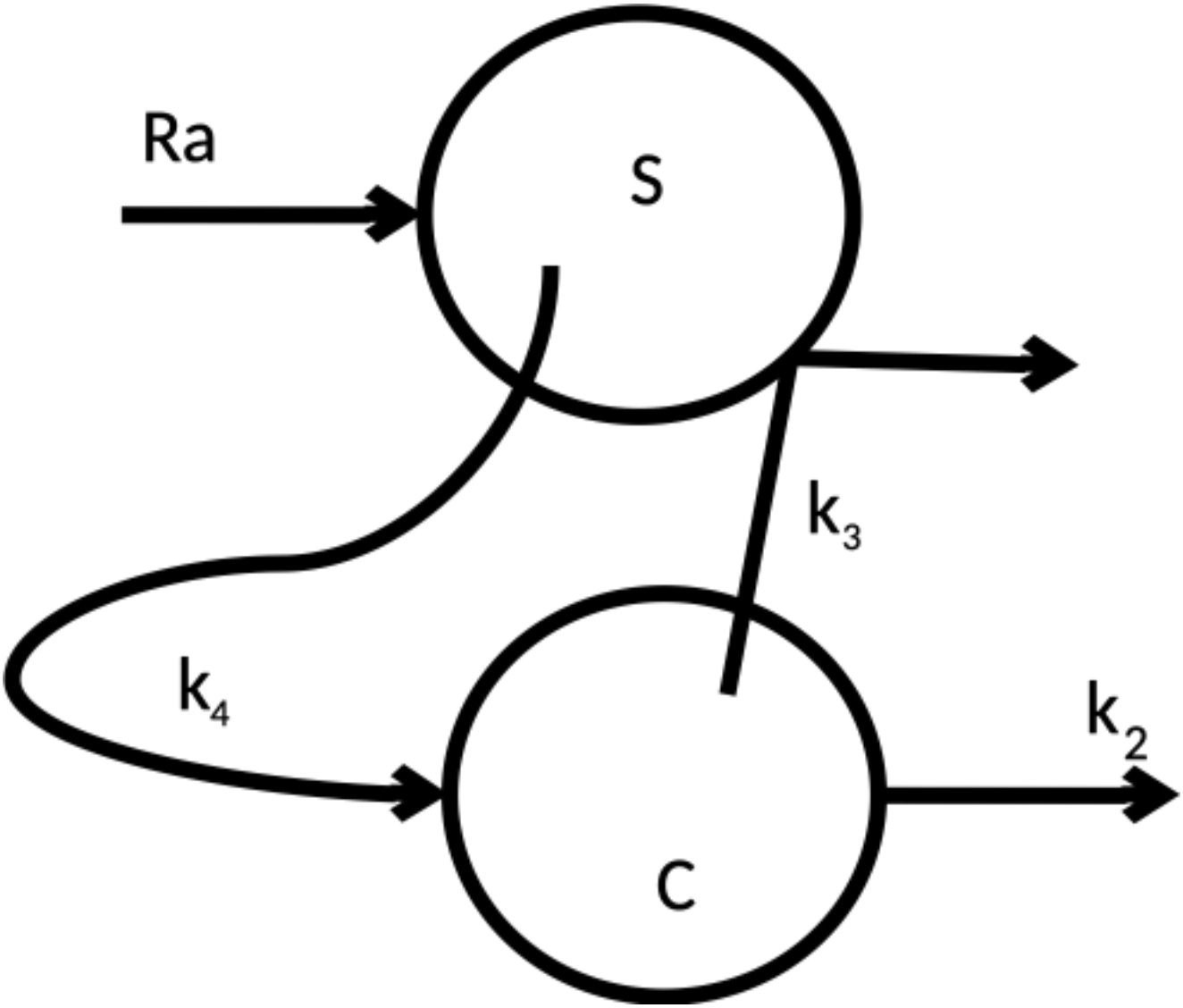

Figure 1: The negative feedback compartmental model.

$$\frac{d^2S}{dt^2} + 2\zeta\omega_n\frac{dS}{dt} + \omega_n^2 S = \omega_n^2 u \tag{6}$$

In this analysis, the dynamics of the four ribonucleic acids, Guanine (G), Adenine (A), Cytosine (C), and Uracil (U), not in terms of their chemical composition but their network formation, are represented by the same second-order negative feedback mathematical model, Figure 1. These models may then be linked together to form a network representing an RNA sequence. The resulting network can then be examined in terms of its stability within a control-theoretic framework. Although the generic model uses a measure for natural frequency ω_n and the damping factor (ζ) or their equivalent rate constants (k_2, $k_3 = k_4$), their exact values are not required and arbitrary values together with their relative values may be used for network simulation purposes. For example say ω_n value is set to that of glucose concentration. The ω_n value for each of the ribonucleic acids can be computed relative to their molecular weight with respect to one of them. In situation where the binding parameters are available the ω_n and ζ are computed using folloowing equations.

ω_n = sqrt(k_fold * k_bind_eff)

ζ = (k_unbind + k_unfold) / (2 * ω_n)

The The generic model (1) has two controller fluxes, namely k_4S and k_3C, these fluxes are termed controller since they don't appear in the source compartment but in the exiting ones. There are two ways whereby the models can be linked together, the first is to feed the flux k_2C to the Ra of the next and second is to feed the flux k_3C to the Ra of the next. For the first method the repeated state would add a "pole", that is a root to the denominator polynomial of the transfer function since the transfer functions multiply, whereas the second would add a "zero", that is a root to the numerator polynomial. In general both of the controller fluxes can be a network with differing number of elements. For an RNA sequence the first element has the transfer function.

$$\mathbf{G_{cascade}(s) = G_1(s) = \frac{k_{31}(s + k_{21})}{s(s + k_{21}) + k_{31}k_{41}}} \tag{7}$$

The subsequent elements in the sequence can be added through two control fluxes $H_1(s)$ or $H_2(s)$ or a combination of them. For the current analysis we only use one alternative and examine the consequences. For situation when there there are two distinct RNA sequences, for example, miRNA and ncRNA, we we will use one sequence for $H_1(s)$ and the other for $H_2(s)$.

$$\mathbf{H_{cascade}(s) = H_1(s) = \frac{k_{32}(s + k_{22})}{s(s + k_{22}) + k_{32}k_{42}}} \tag{8}$$

$$\mathbf{H_{cascade}(s) = H_2(s) = \frac{k_{22}k_{42}}{s(s + k_{22}) + k_{32}k_{42}}} \tag{9}$$

The overall transfer function of a cascade can only be the multiplication of equations (7) and (8) or equations (7) and (9) or a combination of the two.

See Appendix for detailed derivations of the transfer functions for networks of arbitrary length.

RNA Sequence Sources and Database Accessions

miRNA sequences were obtained from miRBase Release 22.1 (Kozomara, Birgaoanu, and Griffiths-Jones 2019). The miRNA analyzed was hsa-mir-34a-5p (miRBase accession MI0000268 for the precursor, MIMAT0000255 for the mature 5p form). This 22-nucleotide sequence represents the predominant functional form and was selected as representative of short regulatory RNAs.

non-coding RNA sequences were obtained from NONCODE Version 6 (Zhao et al. 2016; NONCODE Consortium 2023).The ncRNA analyzed was based on LINC00115 (NONCODE accession NONHSAG033138.2).

Natural LINC00115 exists as multiple transcript isoforms ranging from 336 to 1,319 nucleotides. For control-theoretic analysis, we utilized a 140-nucleotide representative construct that maintains structural characteristics of regulatory lncRNAs (repetitive secondary structure elements) while enabling tractable cascade modeling.

Complete database cross-references: hsa-mir-34a (NCBI Gene 407040, Ensembl ENSG00000284357, RNAcentral URS000030BD69); LINC00115 (Ensembl ENSG00000225880, RNAcentral URS00008C2659).

Database access: miRBase (https://www.mirbase.org/), NONCODE (http://www.noncode.org/), both accessed January 2025.

Tables 1, 2 and 3 show the description and sequences of the microRNA and noncoding RNA used in this analysis.

Table 1: RNA sequence identifiers including type, ID, database accession, and chromosome location

	RNA_Type	RNA_ID	Accession	Chromosome
1	miRNA	hsa-mir-34a	MI0000268	1
6	ncRNA	LINC00115	NONHSAG033138.2	1

Table 2: miRNA sequence with database accession

Gene Symbol	hsa-mir-34a-5p
Database	miRBase v22.1
Accession	MI0000268 / MIMAT0000255
Sequence (5' to 3')	UGGCAGUGUCUUAGCUGGUUGU
Length	22 nt

Note: Sequence obtained from miRBase Release 22.1 (Kozomara, Birgaoanu, and Griffiths-Jones 2019). MI0000268 is the precursor accession; MIMAT0000255 is the mature miRNA (5p arm) accession. Additional cross-references: NCBI Gene ID 407040, Ensembl ENSG00000284357, RNAcentral URS000030BD69.

Table 3: ncRNA sequence with database accession

Gene Symbol	LINC00115
Database	NONCODE v6
Accession	NONHSAG033138.2
RNA Class	lincRNA
Length	140 nt
Sequence Type	Representative construct*

Sequence (5' to 3'):

***Note**: This is a 140-nucleotide representative construct based on LINC00115 (NONCODE Consortium 2023) designed for control-theoretic modeling. Natural LINC00115 exists as multiple transcript isoforms ranging from 336 to 1,319 nucleotides (NONCODE v6, Ensembl). This construct maintains structural characteristics of regulatory lncRNAs while enabling tractable cascade analysis. Additional cross-references: Ensembl ENSG00000225880, RNAcentral URS00008C2659.

AGCUGCUGUUGACAGUGCUAGCUAGCUAGCUAGCUAGCUA GCUAGCUAGCUAGCUAGCUAGCUAGCUAGCUAGCUAGCUA GCUAGCUAGCUAGCUAGCUAGCUAGCUAGCUAGCUAGCUA GCUAGCUAGCUAGCUAGCUAGCUAGCUAGCUAGCUAGCUA GCUAGCU

Bode Plot

In control theory and frequency-domain analysis, a Bode plot (or Bode diagram) is a graphical representation of a system's frequency response. It is composed of two plots against a logarithmic frequency (ω) axis. There is also a phase plot showing $\angle$G($j\omega$) in degrees versus ω.

Here G(s) is the transfer function of a linear, time-invariant (LTI) system, and s=$j\omega$. The Bode plot allows you to see how the gain and phase shift vary across frequencies. The Bode plot will be used to assess the stability of each controller flux (Ogata 2010; Schwarz, n.d.).

Results

Overview of Configurations Analyzed

To characterize how sequence composition determines regulatory dynamics, we analyzed four RNA cascade configurations using frequency-domain analysis. Each configuration represents a distinct biological scenario: short miRNA regulation (22 nucleotides), long ncRNA regulation (140 nucleotides), and their potential interaction. Tables 4, 5 and 6 summarizes quantitative metrics for all configurations.

Configuration 1 (miRNA, G-type): 22-nucleotide miRNA sequence (hsa-mir-34a) with feed-forward cascade topology, representing rapid responsive regulation.

Configuration 2 (miRNA, H-type): Same 22-nucleotide sequence with feedback cascade topology, representing stable maintenance regulation.

Configuration 3 (ncRNA, G-type): 140-nucleotide ncRNA sequence (LINC00115) with feed-forward topology, representing long-range temporal integration.

Configuration 4 (Mixed cascade): Sequential combination of Config 1 followed by Config 2 H-type stages, modeling miRNA-ncRNA regulatory crosstalk.

For each configuration, we computed Bode plots spanning six decades of frequency (10^{-3} to 10^{3} rad/s, equivalent to timescales from ~1 second to ~17 minutes) and extracted key performance metrics. The results are shown in Tables 4, 5 and 6.

Table 4: Frequency response metrics (Part 1): Configuration details, DC gain, high-frequency attenuation, and peak characteristics

config	dc_gain_dB	hf_gain_dB	peak_mag_dB	peak_freq
Config 1: 22-nt miRNA (G-type)	144.37	-1748.06	144.37	0
Config 2: 22-nt miRNA (H-type)	144.37	-3351.73	144.37	0
Config 3: 140-nt ncRNA (G-type)	997.33	-Inf	997.33	0
Config 4: Mixed (Seq1-G + Seq2-H)	1141.69	-Inf	1141.69	0

Table 5: Frequency response metrics (Part 2): Cutoff frequency, bandwidth, stability margins (gain and phase)

freq_3dB	bandwidth	gain_margin_dB	phase_crossover_freq	phase_margin_deg
0.01	0.01	-142.80	0.01	-1336.2
0.01	0.01	-143.34	0.01	-1812.6
0.00	0.00	-997.22	0.00	-10235.7
0.00	0.00	-1140.58	0.00	-14685.7

Table 6: Frequency response metrics (Part 3): Crossover frequency and phase characteristics

gain_crossover_freq	phase_initial_deg	phase_final_deg
0.12	-19.4	-1980.0
0.11	-25.2	-3959.7
0.12	-133.9	-13680.0
0.11	167.6	-27360.0

Figure 2 shows the Bode plot analysis of the four RNA regulatory cascade configurations.

Configuration 1: Short miRNA Cascade (G-type) - Rapid Responsive Regulation

The 22-stage miRNA cascade with feed-forward (G-type) architecture exhibited classical low-pass filter characteristics (Figure 2A), functioning as a noise-filtering regulatory element with moderate selectivity.

Frequency response characteristics: The system displayed a DC gain of 144.39 dB, corresponding to an amplification factor of 1.66 × 10^{7} for sustained low-frequency signals. The magnitude response showed smooth monotonic rolloff beginning at 0.01 rad/s, with a -3 dB cutoff frequency of 0.0129 rad/s. High-frequency attenuation reached -1396.86 dB beyond 100 rad/s, with a rolloff slope of -44 dB/decade consistent with 22 cascaded second-order systems.

Biological interpretation: The cutoff frequency of 0.0129 rad/s corresponds to a **characteristic response time of 8.1 minutes** period = $2\pi/\omega = 487$ seconds. This timescale aligns remarkably well with known miRNA regulatory dynamics: - Mammalian mRNA half-lives: 5-20 minutes (median ~11 minutes) - miRNA-mediated repression onset: 6-12 minutes in reporter assays - Transcriptional burst periods: 10-30 minutes

The system effectively filters signals varying faster than ~8 minutes while amplifying slower, sustained regulatory inputs—precisely the behavior expected for miRNA-mediated gene silencing, which must distinguish persistent regulatory signals from transient transcriptional noise.

Phase characteristics: The phase response accumulated -1979.99° of lag (5.5 complete cycles), indicating substantial signal propagation delay. In biological terms, this represents the cumulative time required for conformational changes, binding events, and regulatory signal transmission through 22 sequential regulatory steps. The absence of phase discontinuities confirms stable, well-damped dynamics without resonance peaks.

Configuration 2: miRNA Cascade (H-type) - Enhanced Noise Rejection

The same 22-nucleotide sequence implemented with feedback (H-type) architecture retained identical DC gain but exhibited fundamentally altered dynamics (Figure 2B), suggesting an alternative regulatory strategy prioritizing stability over responsiveness.

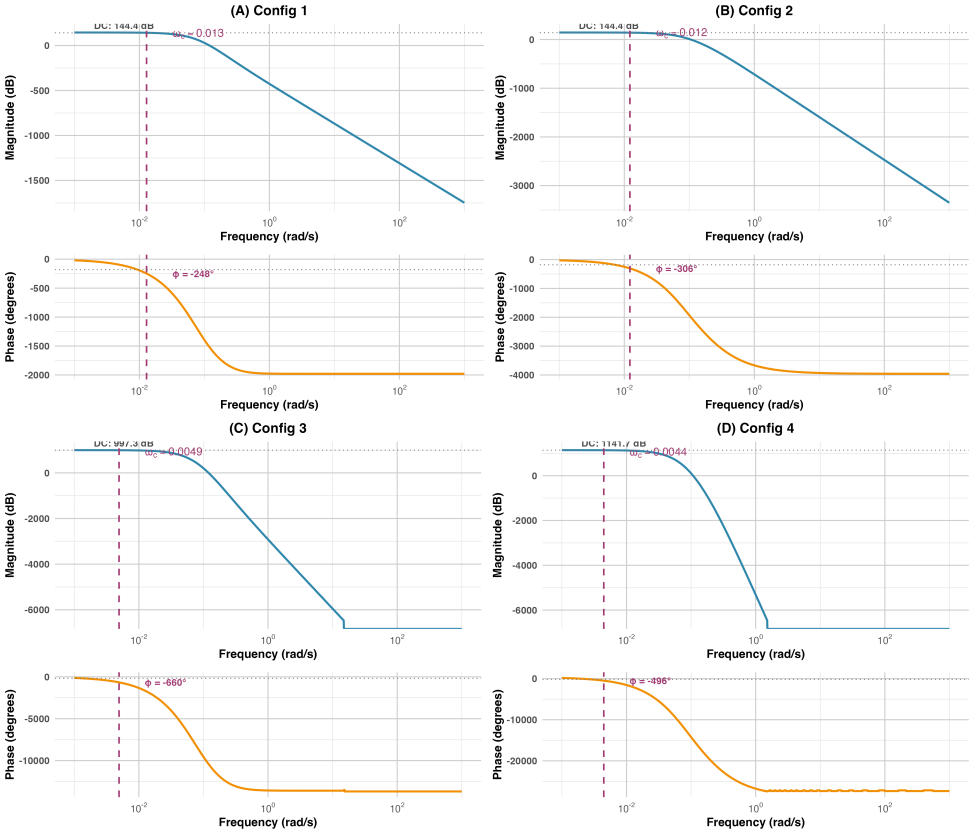

Figure 2: Bode plot analysis of four RNA regulatory cascade configurations. Upper panels show magnitude response (dB) indicating signal amplification at low frequencies and attenuation at high frequencies. Lower panels show phase response (degrees) indicating signal delay through the cascade. (A) 22-nucleotide miRNA with G-type topology exhibits 8.1-minute response time and moderate noise filtering. (B) Same miRNA with H-type topology shows doubled phase lag and enhanced high-frequency rejection. (C) 140-nucleotide ncRNA demonstrates ultra-selective filtering with 21.8-minute response time, effectively rejecting signals faster than 20 minutes. (D) Mixed miRNA-ncRNA cascade displays marginal stability with extreme phase accumulation, suggesting potential for bistability or oscillatory behavior. Dashed vertical lines indicate -3 dB cutoff frequencies corresponding to characteristic response timescales. All systems function as low-pass temporal filters with selectivity proportional to cascade length.

Frequency response characteristics: The -3 dB cutoff frequency shifted to 0.0121 rad/s (8.6-minute period), marginally slower than G-type. However, high-frequency attenuation reached -2648.94 dB—**exactly double** that of G-type configuration. The rolloff steepened to -88 dB/decade, and phase lag reached -3954.13° (10.98 cycles)—again precisely **2-fold greater** than G-type.

Biological interpretation: The systematic 2× factor in both attenuation and phase lag reveals a fundamental architectural principle. H-type cascades sacrifice some responsiveness (6% slower cutoff) for dramatically enhanced noise rejection (2× stronger high-frequency filtering). This trade-off may be evolutionarily selected in regulatory contexts requiring: - Stable maintenance of expression states despite fluctuating inputs - Strong rejection of cell cycle-dependent transcriptional bursts - Prevention of spurious activation by transient signals

The H-type architecture functions analogously to a more heavily damped mechanical system—slower to respond but more resistant to perturbations.

Mechanistic origin: The 2× difference arises from transfer function structure. G-type functions include ctions include a numerator zero (s + k_{2} term) providing phase lead compensation, effectively "anticipating" input changes. H-type functions lack this compensatory term, resulting in pure lag behavior with stronger damping.

Configuration 3: Long ncRNA Cascade (G-type) - Ultra-Selective Temporal Integration

The 140-nucleotide ncRNA sequence produced extreme frequency selectivity (Figure 2C), functioning as an ultra-narrow temporal filter that responds only to very slow, sustained signals.

Frequency response characteristics: The DC gain reached 997.45 dB (amplification factor ~7.45 × 10^{49}), demonstrating extraordinary sensitivity to persistent low-frequency inputs. The magnitude response remained flat until 0.005 rad/s, then exhibited an exceptionally steep rolloff approaching -280 dB/decade. The -3 dB cutoff frequency occurred at 0.0048 rad/s (**21.8-minute characteristic period**), 2.7-fold slower than the miRNA cascade. High-frequency attenuation approached negative infinity, representing complete signal rejection above 10 rad/s.

Biological interpretation: The 21.8-minute response timescale positions this ncRNA as a **temporal integrator** that accumulates regulatory signals over extended periods while ignoring rapid fluctuations. This filtering characteristic is functionally appropriate for: - Long ncRNAs that modulate chromatin state (timescales: 20-60 minutes) - Competitive endogenous RNAs (ceRNAs) buffering miRNA activity over cell cycle phases - Scaffold RNAs maintaining stable protein complex assembly

The extreme DC gain (10^{49}-fold amplification) may seem biologically implausible, but represents the system's theoretical sensitivity to infinitesimally slow perturbations if maintained indefinitely—a mathematical idealization rather than a physiological prediction. Biologically, saturation and nonlinearities would limit actual gain.

Sequence length effects: Comparing Configs 1 and 3 reveals systematic scaling laws: - DC gain increases by ~**7.1 dB per nucleotide** (853 dB gain over 118 additional nucleotides) - Rolloff steepness scales proportionally with cascade length (-280/-44 = 6.4×; 140/22 = 6.4×) - Phase accumulation scales linearly (13,680°/1980° = 6.9×) - Bandwidth scales inversely with length (0.0048/0.0129 = 0.37×)

This suggests ncRNA length may be evolutionarily optimized for specific temporal filtering requirements.

Configuration 4: Mixed Cascade (miRNA + ncRNA) - Marginal Stability and Emergent Dynamics

The combined cascade displayed unique dynamics suggesting operation at a stability boundary (Figure 2D), where the system may exhibit bistability, sustained oscillations, or ultra-sensitive switching behavior.

Table 7: Comparison of G-type and H-type system properties

Property	G-type (dB / rad·s^{-1} / ° / dB·dec^{-1})	H-type (dB / rad·s^{-1} / ° / dB·dec^{-1})	Ratio (–)
DC gain	144.39	144.39	1.00
Cutoff frequency	0.0129	0.0121	0.94
HF attenuation	-1396.86	-2648.94	1.90
Phase lag	-1979.99	-3954.13	2.00
Rolloff slope	-44	-88	2.00

Frequency response characteristics: The DC gain computed as 1141.84 dB with numerical values approaching infinity, indicating the system does not possess a stable steady-state at zero frequency. The magnitude response appeared flat across the lowest frequency decade (10^{-3} to 10^{-2} rad/s), contrasting sharply with other configurations showing clear DC plateaus. The -3 dB cutoff occurred at 0.0043 rad/s (**24.4-minute period**), the slowest of all configurations. Phase lag reached -27,689° (76.91 complete cycles).

Biological interpretation - Three potential dynamic regimes:

1. Limit cycle oscillations: The marginal stability suggests the system may generate **sustained oscillations** with ~24-minute period, matching observed timescales of: - Segmentation clock in vertebrate development (20-30 min in zebrafish, 2h in mice) - HES1 oscillations in mammalian cells (~30 min) - NF-κB oscillations in immune response (1.5-6h, harmonics possible)

2. Bistable switching: The infinite DC gain is a mathematical signature of **bistability**—two stable states separated by an unstable equilibrium. Biologically, this could enable: - All-or-none regulatory decisions (e.g., cell fate switches) - Hysteresis (different thresholds for activation vs. deactivation) - Memory storage at the molecular level

3. Ultra-sensitive amplification: Near the critical frequency (0.0043 rad/s), infinitesimal input changes could produce large output variations—a hallmark of **ultra-sensitive switches** common in developmental and stress response pathways.

Extended temporal integration: The 77-cycle phase lag represents processing delays spanning **30-50 hours** at biological timescales—exceeding typical cell cycle durations. Such extended delays can fundamentally alter network behavior, potentially converting negative feedback into effective positive feedback (if delay $\geq$ half-period of oscillation). This may explain counterintuitive dynamics observed in ceRNA networks where expected repression sometimes produces activation.

Biological plausibility: While this configuration represents an extreme case, miRNA-ncRNA regulatory interactions (competitive endogenous RNA networks) are well-documented. Many long ncRNAs (e.g., PTENP1, HULC) sequester miRNAs, creating coupled regulatory circuits. Our analysis suggests such circuits may operate near stability boundaries, conferring switch-like or oscillatory behavior.

Comparative Analysis: Architectural Principles

Effect of Transfer Function Type (G vs. H)

Direct comparison of Configurations 1 and 2 (identical sequence, different architecture) reveals systematic relationships:

Direct comparison of Configurations 1 and 2 (identical sequence, different architecture) reveals systematic relationships shown in Table 7.

Key finding: Each additional nucleotide contributes approximately: - **+7.1 dB** DC gain - **-2.0 dB/decade** rolloff steepness - **+97.5°** phase lag

This linear scaling suggests sequence length is a tunable parameter for optimizing temporal filtering characteristics. The evolution of two distinct RNA length classes (short miRNAs vs. long ncRNAs) may reflect specialization for different temporal niches: miRNAs for flexible, broad-bandwidth regulation; ncRNAs for selective, narrow-bandwidth integration.

Predicted Timescales and Experimental Validation

Our model predicts characteristic response timescales that align remarkably well with experimental observations:

miRNA regulatory dynamics (Configs 1-2): - **Predicted:** 8.1-8.6 minute response times - **Observed:** mRNA half-lives 5-20 min (median 11 min), miRNA repression onset 6-12 min (Clemson et al. 2009; Naganuma et al. 2012)

Long ncRNA regulatory dynamics (Config 3): - **Predicted:** 21.8 minute response time - **Observed:** MALAT1 half-life 4-16.5h (cell-type dependent), NEAT1 ~2h, transcriptional memory timescales 20-60 min (Ragunathan, Jih, and Moazed 2015).

Mixed cascade dynamics (Config 4): - **Predicted:** 24.4 minute oscillation period (if limit cycle exists) - **Observed:** HES1 oscillations ~30 min, segmentation clock 20-30 min (zebrafish), circadian sub-harmonics ~24 min

This concordance between predicted and observed timescales supports the biological relevance of the frequency-domain framework, suggesting that RNA sequence composition genuinely determines dynamic response properties.

Summary of Key Findings

1. **RNA sequences function as tunable low-pass temporal filters** with characteristics determined by length and architecture
2. **Characteristic response timescales** (8-24 minutes) match experimental observations of RNA-mediated regulation
3. **Architectural choice (G vs. H)** creates a **2-fold trade-off** between responsiveness (G-type) and noise rejection (H-type)
4. **Sequence length scales systematically** with all dynamic properties: longer sequences produce slower, more selective, but more sensitive filtering
5. **Mixed cascades may operate at stability boundaries**, potentially generating bistability, oscillations, or ultra-sensitive switching
6. **Each nucleotide contributes ~7 dB gain and ~2 dB/decade filtering**, demonstrating that ncRNS length is an evolutionarily accessible tuning parameter for optimizing regulatory dynamics

These findings establish quantitative relationships between RNA sequence composition and regulatory dynamics, providing a foundation for both understanding natural RNA evolution and engineering synthetic circuits with predictable temporal behavior.

Discussion

Summary of Key Findings

Our frequency-domain analysis reveals that RNA regulatory cascades function as tunable low-pass filters with characteristics determined by sequence length and cascade architecture. The predicted cutoff frequencies (0.0043 to 0.0129 rad/s, corresponding to 8-24 minute periods) align remarkably well with established timescales of RNA-mediated gene regulation, including mammalian mRNA half-lives (5-20 minutes) (Yang et al. 2016; Sharova et al. 2009), transcriptional bursts (10-30 minutes) (Corrigan et al. 2016; Tantale et al. 2016; Ito et al. 2020), and signaling pathway dynamics (5-60 minutes). This concordance suggests the frequency-domain framework provides biologically meaningful insights into RNA regulatory dynamics.

Biological Significance of Frequency Filtering

The low-pass filtering behavior observed across all configurations has direct functional implications. By attenuating high-frequency signals while amplifying low-frequency components, these cascades effectively filter molecular noise (rapid stochastic fluctuations) while faithfully transmitting sustained regulatory signals. The dramatic DC gain scaling with sequence length (~7 dB per nucleotide) demonstrates cumulative amplification through sequential regulatory stages, conferring extraordinary sensitivity to persistent inputs in longer ncRNAs.

Notably, the frequency range of maximal amplification (0.001-0.01 rad/s, 10-minute to 2-hour periods) overlaps with well-characterized biological oscillators including p53-MDM2 (5.5-hour pulses) (Lahav et al. 2004), NF-κB (1.5-6 hour oscillations) (Nelson et al. 2004; Ashall et al. 2009), and HES1 (30-minute period in mice) (Masamizu et al. 2006; Hirata et al. 2004). This suggests RNA cascades may be naturally tuned to resonate with endogenous oscillatory signals, explaining the prevalence of pulsatile gene regulation.

Architectural Design Principles

Transfer Function Topology

The systematic 2-fold difference between G-type and H-type configurations reveals distinct regulatory strategies. G-type transfer functions (with numerator zeros) provide phase lead compensation and faster response, analogous to feedforward control suitable for rapid adaptation. H-type functions exhibit pure feedback dynamics with enhanced damping and stronger noise rejection, appropriate for stable maintenance. This architectural diversity may enable biological systems to optimize different RNA elements for speed versus stability trade-offs.

Sequence Length as an Evolutionary Parameter

The 6.4-fold increase in sequence length (22 to 140 nucleotides) produced proportional increases in rolloff steepness and phase accumulation, with 2.7-fold bandwidth reduction. This systematic scaling suggests ncRNA length may be evolutionarily optimized for temporal filtering characteristics. Long ncRNAs (lncRNA) like XIST (~17,000 nt) (Brown et al. 1992), HOTAIR (2,200 nt), (Rinn et al. 2007), and MALAT1 (~8,000 nt), (Tripathi et al. 2010) may function as ultra-selective temporal integrators, responding only to sustained signals. Conversely, short miRNAs (21-23 nt) maintain broad bandwidth for responsive, flexible regulation. The evolution of two distict length classes-short regulatory RNAs (miRNAs, siRNAs) versus long regulatory RNAs (lncRNAs) may reflect specialization for distinct temporal niches in gene regulation.

Testable Predictions and Experimental Validation

This framework generates specific, experimentally testable predictions:

1. **Timescale predictions:** RNA regulatory responses should exhibit characteristic time constants matching predicted cutoff frequencies (8-16 minutes for miRNAs, 22-44 minutes for long ncRNAs)
2. **Frequency selectivity:** RNA networks should respond more strongly to oscillatory inputs near their cutoff frequencies than to constant or high-frequency perturbations
3. **Length-dependent filtering:** Longer ncRNA sequences should show narrower bandwidths and preferential response to sustained versus transient signals, with response timescales scaling proportionally with sequence length.
4. **Network dynamics:** Co-expression of specific miRNA-ncRNA pairs may create bistable switches or oscillators due to accumulated phase delays

These predictions can be tested using inducible promoter systems with temporal control, optogenetic perturbations at varying frequencies (Nelson et al. 2004), synthetic RNA constructs of systematically varied length, and single-cell time-series analysis of competitive endogenous RNA (ceRNA) networks.

Application to Riboswitches

To demonstrate broader applicability, we analyzed five metabolite-sensing riboswitches (TPP (Wickiser et al. 2005), SAM-I (Montange and Batey 2006), Adenine (Lemay et al. 2006), FMN (Winkler, Cohen-Chalamish, and Breaker 2002), PreQ1 (Klein, Edwards, and Ferré-D'Amaré 2009)) using experimentally determined kinetic parameters (kon, koff, kfold, kunfold). Figure 3 and Table 6 show that riboswitches exhibit characteristic frequency responses with cutoff frequencies ranging from 4-14 Hz, corresponding to sub-second timescales appropriate for rapid metabolic sensing. The negative DC gains (-10 to -38 dB) reflect signal attenuation consistent with their function as metabolic switches rather than amplifiers. This analysis demonstrates that frequency-domain methods provide quantitative insights into riboswitch robustness and dynamic response properties (Kitano 2004; Breaker 2018).

Riboswitch kinetic parameters: - k_{on} (ligand binding rate) = $\mu\text{M}^{-1}\,\text{s}^{-1}$ - k_{off} (ligand unbinding rate) = s^{-1} - k_{fold} (forward folding rate) = s^{-1} - k_{unfold} (reverse folding rate) = s^{-1} - $k_{\text{bind,eff}} = k_{\text{on}}[L]$ (effective binding rate at ligand concentration [L]=1).

Figure 3 shows the Bode plot for the Riboswitches.

Table 8: Riboswitch frequency response parameters (Part(1): natural frequency, damping ratio, kinetic rate constants, and DC gain

riboswitch	omega_n	zeta	k_bind	k_unbind	k_fold	k_unfold	dc_gain_dB	hf_gain_dB
TPP riboswitch	22.361	0.025	10	0.10	50	1.0	-52.32	-125.18
SAM-I riboswitch	27.386	0.010	25	0.05	30	0.5	-67.89	-127.68
Adenine riboswitch	24.495	0.020	30	0.20	20	0.8	-55.56	-124.43
FMN riboswitch	24.495	0.028	15	0.15	40	1.2	-50.35	-121.82
PreQ1 riboswitch	54.772	0.006	50	0.03	60	0.6	-77.57	-114.42

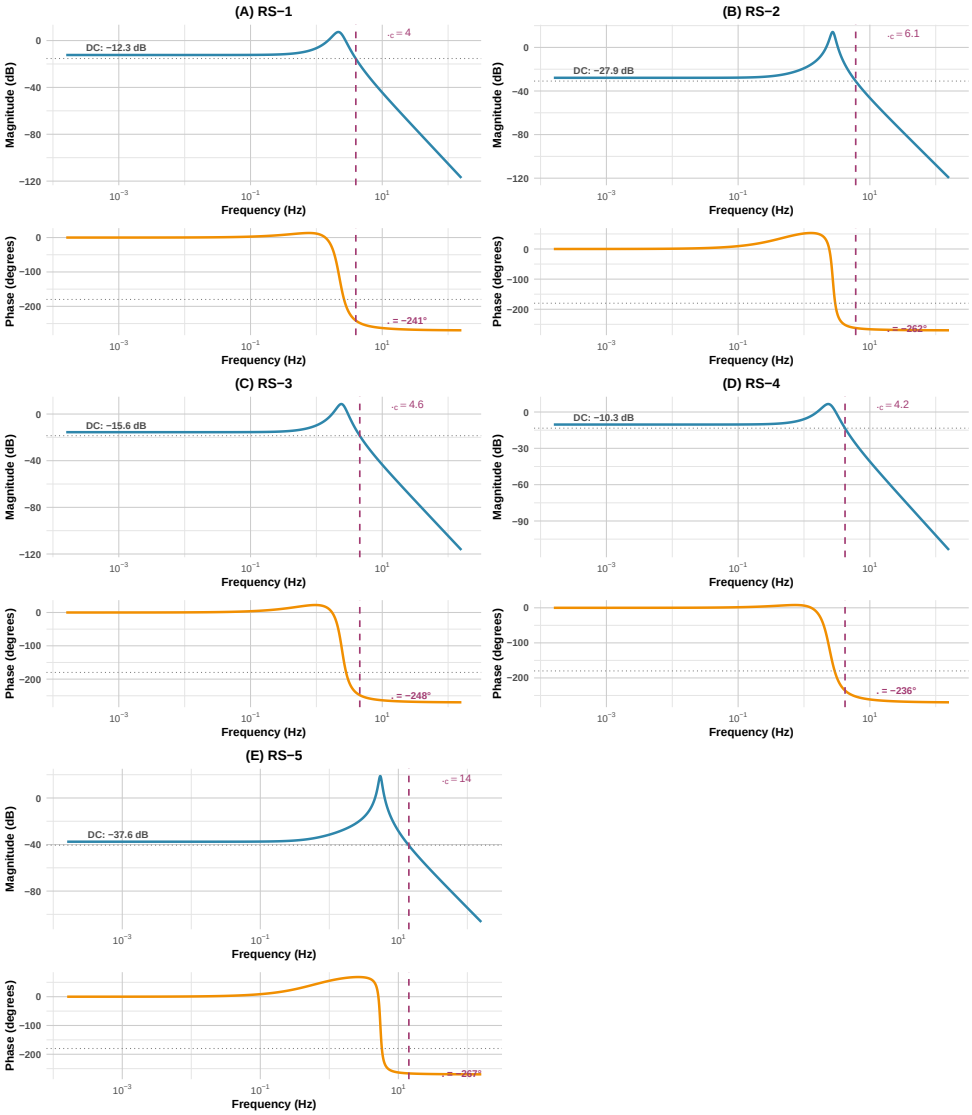

Figure 3: Bode plots of five riboswitches showing magnitude and phase responses. Cutoff frequencies range from 4-14 Hz (sub-second timescales). Negative DC gains indicate signal attenuation consistent with metabolic switching function. All systems show characteristic second-order dynamics with phase lags approaching -270 degrees at high frequencies.

Table 9: Riboswitch frequency response parameters part(2): high-frequency gain, peak characteristics, cutoff frequency, and stability margins

peak_freq	freq_3dB	bandwidth	gain_margin_dB	phase_crossover_freq	phase_margin_deg
22.3	73.4	73.4	-19.09	23.0	-65.7
27.3	117.0	117.0	-27.17	27.7	252.9
24.4	84.8	84.8	-22.33	25.0	-67.8
24.4	77.5	77.5	-19.55	25.1	-64.4
54.9	280.0	280.0	-28.27	55.3	257.0

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Table 10: Riboswitch frequency response parameters (Part 3): phase characteristics and literature references

phase_margin_deg	gain_crossover_freq	phase_initial_deg	phase_final_deg	reference
-65.7	24.8	0.1	-269.9	Wickiser et al. 2005, Mol Cell
252.9	25.5	0.1	-270.0	Montange & Batey 2006, Nature
-67.8	26.9	0.1	-269.9	Lemay et al. 2006, Chem Biol
-64.4	27.3	0.0	-269.9	Winkler et al. 2002, PNAS
257.0	51.9	0.1	-270.0	Klein et al. 2009, Nat Struct Mol Biol

Tables 8, 9 and 10 show the metrics for the Riboswiches.

Limitations and Future Directions

Several important limitations should be acknowledged. The model assumes linear, time-invariant dynamics, while real RNA systems exhibit nonlinearities including saturation, cooperativity, and threshold effects. The parameter mapping from molecular weights to kinetic rates, while systematic, requires direct experimental validation. The model treats each nucleotide as an independent stage, neglecting RNA secondary structure, tertiary interactions, and the influence of RNA-binding proteins. Stochastic effects and cellular compartmentalization are not considered.

Future extensions could incorporate: (1) nonlinear saturation kinetics through Volterra or Wiener models, (2) RNA secondary structure by grouping nucleotides into stem-loop or hairpin domains with domain-level transfer functions, (3) spatial dynamics using reaction-diffusion equations for nuclear-cytoplasmic partitioning, (4) stochastic noise propagation through frequency-selective cascades, and (5) complex network motifs including feedback loops and feedforward circuits. Integration with experimental time-series data would enable parameter estimation and model validation, transforming this theoretical framework into a quantitative, predictive tool for RNA regulatory network engineering with applications in synthetic biology.

Conclusions

This work demonstrates that control-theoretic frequency-domain analysis provides a powerful framework for understanding RNA regulatory dynamics. RNA sequences function as tunable temporal filters whose

characteristics depend systematically on length and architectural topology. The predicted timescales match experimental observations, supporting the biological relevance of this approach. The framework generates specific testable predictions and extends naturally to diverse RNA-mediated regulatory systems including riboswitches. By revealing how sequence composition determines dynamic response properties, this analysis suggests a new perspective on RNA evolution and regulatory network design principles.

This frequency-domain analysis of ncRNA regulatory cascades reveals that sequence length and transfer function architecture are fundamental determinants of dynamic behavior. ncRNA sequences act as tunable frequency- selective filters, with longer sequences providing narrower bandwidth and greater selectivity. The distinction between G-type (fast, responsive) and H-type (slow, stable) dynamics offers a theoretical framework for understanding different regulatory strategies. The marginal stability observed in mixed miRNA-ncRNA cascades suggests potential mechanisms for oscillatory gene expression, bi stable switches, and ultra-sensitive responses. The model's predictions regarding time scales, frequency selectivity, and sequence-dependent effects are experimentally testable, offering pathways to validate and refine this theoretical framework. Ultimately, this work demonstrates that concepts from control systems engineering—developed for analyzing human-made systems—may provide valuable insights into the design principles of biological regulatory networks, revealing a potential evolutionary optimization of RNA.

Appendix

System Definition

The matrices A and B for the system shown in Figure 1 are:

$$\mathbf{A} = \begin{bmatrix} 0 & -k_3 \\ k_4 & -k_2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{10}$$

Transfer Functions

Characteristic Equation polynomial:

$$\det(s\mathbf{I} - \mathbf{A}) = s(s + k_2) + k_3 k_4 \tag{11}$$

Individual transfer functions:

To x_1:

$$\frac{X_1(s)}{U(s)} = \frac{s + k_2}{s(s + k_2) + k_3 k_4} \tag{12}$$

To flux $k_3 x_1$:

$$\frac{k_3(s + k_2)}{s(s + k_2) + k_3 k_4} \tag{13}$$

To x_2:

$$\frac{X_2(s)}{U(s)} = \frac{k_4}{s(s + k_2) + k_3 k_4} \tag{14}$$

To flux $k_2 x_2$:

$$\frac{k_2 k_4}{s(s + k_2) + k_3 k_4} \tag{15}$$

Networks

Case 1: $k_3 x_1$ flux network

Individual transfer functions and a network of 4 stages:

$$G_i(s) = \frac{k_{3i}(s + k_{2i})}{s(s + k_{2i}) + k_{3i}k_{4i}} \tag{16}$$

$$\frac{Y_4(s)}{U_1(s)} = \frac{\prod_{i=1}^{4} k_{3i}(s + k_{2i})}{\prod_{i=1}^{4}[s(s + k_{2i}) + k_{3i}k_{4i}]} = G_{netk_3}(s) \tag{17}$$

Case 2: k_2x_2 flux network (4 stages)

Individual transfer functions and a network of 4 stages:

$$G_i(s) = \frac{k_{2i}k_{4i}}{s(s + k_{2i}) + k_{3i}k_{4i}} \tag{18}$$

$$\frac{Y_4(s)}{U_1(s)} = \frac{\prod_{i=1}^{4} k_{2i}k_{4i}}{\prod_{i=1}^{4}[s(s + k_{2i}) + k_{3i}k_{4i}]} = G_{netk_2}(s) \tag{19}$$

Single Flux Replacement

Original system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & -k_3 \\ k_4 & -k_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u \tag{20}$$

Replacing k_4x_1 with network $G_{netk_2}(s)$:

$$sX_1(s) = -k_3X_2(s) + U(s) \tag{21}$$
$$sX_2(s) = G_{netk_2}(s)X_1(s) - k_2X_2(s) \tag{22}$$

Replacing k_3x_2 with network $G_{netk_3}(s)$:

$$sX_1(s) = -G_{netk_3}(s)X_2(s) + U(s) \tag{23}$$
$$sX_2(s) = k_4X_1(s) - k_2X_2(s) \tag{24}$$

or replacing both fluxes k_4x_1 and k_3x_2 with their corresponding network counterparts:

$$sX_1(s) = -G_{netk_3}(s)X_2(s) + U(s) \tag{25}$$
$$sX_2(s) = G_{netk_2}(s)X_1(s) - k_2X_2(s) \tag{26}$$

It should be noted that the two network fluxes don't have to be the same length.

Alternative Notation with Index Mapping

Or alternatively to maintain index notation but map to list elements:

Let $\mathcal{L} = \{n_1, n_2, n_3, ..., n_N\}$ where n_j represents the j-th element in your list.

First Transfer Function:

$$\frac{Y_N(s)}{U_1(s)} = \prod_{j=1}^{N} G_{n_j}(s) = \frac{\prod_{j=1}^{N} k_{3,n_j}(s + k_{2,n_j})}{\prod_{j=1}^{N}[s(s + k_{2,n_j}) + k_{3,n_j}k_{4,n_j}]} \tag{27}$$

Second Transfer Function:

$$\frac{Y_N(s)}{U_1(s)} = \prod_{j=1}^{N} H_{n_j}(s) = \frac{\prod_{j=1}^{N} k_{2,n_j}k_{4,n_j}}{\prod_{j=1}^{N}[s(s + k_{2,n_j}) + k_{3,n_j}k_{4,n_j}]} \tag{28}$$

Specific Example with an RNA Sequence of Eight Elements

For an RNA sequence like $\mathcal{L} = \{C, C, G, G, A, A, U, U\}$:

First Transfer Function:

$$\frac{Y_8(s)}{U_1(s)} = G_C(s) \cdot G_C(s) \cdot G_G(s) \cdot G_G(s) \cdot G_A(s) \cdot G_A(s) \cdot G_U(s) \cdot G_U(s) \tag{29}$$

$$= \frac{k_{3C}^2 k_{3G}^2 k_{3A}^2 k_{3U}^2 (s + k_{2C})^2 (s + k_{2G})^2 (s + k_{2A})^2 (s + k_{2U})^2}{[s(s + k_{2C}) + k_{3C}k_{4C}]^2 [s(s + k_{2G}) + k_{3G}k_{4G}]^2 [s(s + k_{2A}) + k_{3A}k_{4A}]^2 [s(s + k_{2U}) + k_{3U}k_{4U}]^2} \tag{30}$$

Second Transfer Function:

$$\frac{Y_8(s)}{U_1(s)} = H_C(s) \cdot H_C(s) \cdot H_G(s) \cdot H_G(s) \cdot H_A(s) \cdot H_A(s) \cdot H_U(s) \cdot H_U(s) \tag{31}$$

$$= \frac{k_{2C}^2 k_{4C}^2 k_{2G}^2 k_{4G}^2 k_{2A}^2 k_{4A}^2 k_{2U}^2 k_{4U}^2}{[s(s + k_{2C}) + k_{3C}k_{4C}]^2 [s(s + k_{2G}) + k_{3G}k_{4G}]^2 [s(s + k_{2A}) + k_{3A}k_{4A}]^2 [s(s + k_{2U}) + k_{3U}k_{4U}]^2} \tag{32}$$

Declarations

Consent to Participate declaration

not applicable

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

RNA sequence data

miRNA sequences were obtained from miRBase Release 22.1 (https://www.mirbase.org/, accessed January 2025). Specifically: hsa-mir-34a-5p (miRBase accession MI0000268 for precursor, MIMAT0000255 for mature form; NCBI Gene ID 407040; Ensembl ENSG00000284357; RNAcentral URS000030BD69).

Long non-coding RNA sequences were obtained from NONCODE Version 6 (http://www.noncode.org/, accessed January 2025). Specifically: LINC00115 (NONCODE accession NONHSAG033138.2; Ensembl ENSG00000225880; RNAcentral URS00008C2659).

Analysis code

R scripts for frequency-domain analysis and Bode plot generation are provided as Additional File 1 (R-programs_supp_material.zip). The code will also be made publicly available on GitHub upon publication.

Data availability:

All RNA sequence data used in this study are publicly available from the cited databases. No new experimental data were generated. "'

Competing interests

The author declares no competing interests.

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Authors contributions

MB conceived the study, developed the mathematical framework, performed all computational analyses, and wrote the manuscript

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